

ASSOCIATION RULE MINING

DEFINITION

- A technique in data mining in order to identify meaningful relationships or patterns between items in large datasets.
- help to uncover how the occurrence of one item or event is related to another within a dataset.
 - useful in domains like healthcare, marketing, and recommendation systems.



PURPOSE

- Discover hidden patterns or relationships in large datasets
- Reveal customer behavior trends
- Support decision-making in various domains
- e.g.: If a patient has high blood sugar and high BMI, they are likely to have hypertension

APRIORI ALGORITHM

- Starts by finding frequent items that appear alone in transactions (count items frequencies)
- Expands to larger item combinations step by step (e.g., pairs, triples)
- Eliminates combinations that don't meet the minimum support value
- Builds rules from the frequent patterns based on how confidently one item predicts another
- Apply confidence and lift to select strong, useful rules

DATASET

Patient ID	Symptoms/Diagnosis
P1	Fever, Cough, Flu
P2	Fever, Sore Throat
P3	Cough, Fatigue, Flu
P4	Fever, Cough, Headache, Flu
P5	Cough, Headache

KEY METRICS



Support:
Indicates how frequently the rule appears in the dataset.

Formula:
$$\text{Support}(X \rightarrow Y) = \frac{\text{(No. of transactions containing X and Y)}}{\text{(Total transactions)}}$$

Example:
$$\text{Support}(\{\text{Fever, Cough}\}) = 3/6 = 0.5$$



Confidence:
Probability of Y given X.

Formula:
$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Example:
$$\text{Confidence}(\{\text{Fever}\} \rightarrow \{\text{Cough}\}) = 0.5 / 0.67 = 0.75$$



Lift:
Indicates strength of a rule over random chance.

Formula:
$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

Example:
$$\text{Lift}(\{\text{Fever}\} \rightarrow \{\text{Cough}\}) = 0.75 / 0.67 \approx 1.12$$

REAL-WORLD APPLICATION: HEALTHCARE DIAGNOSTICS



Doctors and hospitals use association rules to analyze patient symptoms and medical histories, helping to predict diseases early, suggest treatments, or flag high-risk conditions.

- Example:**
- Patients with “fever” and “cough” often are diagnosed with “flu.”
 - Rule: {Fever, Cough} → {Flu}

If a patient shows both symptoms, there's a significantly higher chance they have flu.

Frequent Itemsets (Min Support = 40%)

- {Fever} → 3/5
- {Cough} → 4/5
- {Fever, Cough} → 2/5
- {Fever, Cough, Flu} → 2/5
- {Cough, Flu} → 3/5



Rule	Support	Confidence	Lift
Fever, Cough →	40%	100 %	1,25
Cough → Flu	60%	75 %	1,07
Fever → Cough	60%	66,7%	0,83